

OrangeHRM Functional Testing Planner

Fields	Details
Project Type	HR Management System Functional Testing Capstone
Selected Website	OrangeHRM
Website URL	https://opensource-demo.orangehrmlive.com/
Automation stack	Playwright with JavaScript
Minimum requirement	8 services/modules, 15 test cases per service, total 120 test cases
Testing focus	Authentication, employee management, leave, recruitment, attendance, reporting, attendance, and time tracking.

Project Overview

This capstone project focuses on testing the OrangeHRM web application from the perspective of a QA engineer at an enterprise level. The goal is to analyze employee-management workflows, validate HR-related functionalities, write structured test cases, automate important flows using Playwright with JavaScript, and package the final results in a professional submission-ready format.

Since OrangeHRM contains modules such as authentication, employee management, leave handling, recruitment, attendance, and time tracking, it provides sufficient functional coverage for a complete automation capstone project with at least 120 test cases.

Why OrangeHRM works well for this capstone

- It is a public HR management application accessible through a browser.
- It contains multiple enterprise-level modules suitable for real-world QA testing.
- It supports realistic workflows such as login, employee management, leave approval, and recruitment.
- Playwright with JavaScript is suitable for browser automation and reporting.
- Dummy employee and HR data can be used safely without affecting production systems.

Project objectives

- Write a complete service-wise test plan for the selected modules/services.
- Prepare at least 120 test cases across eight modules.
- Cover positive flows, negative paths, form validation, navigation checks, and simple regression scenarios.
- Automate the chosen flows using Playwright with JavaScript.
- Collect reports, screenshots, and traces so the final submission includes visible proof of execution.

Scope of Testing

The scope stays within the public OrangeHRM employee-management experience. It is intentionally limited to normal HR workflows and avoids anything that depends on backend database modifications, production employee records, or advanced enterprise integrations.

In Scope	Out of Scope
Login, logout, authentication validation, employee management, leave handling, attendance tracking, recruitment workflows, admin management, profile management, and navigation validation	Database modifications, performance testing, security testing, penetration testing, backend API manipulation, production employee records, and third-party integrations

Services/ Modules as Test cases

No.	Service/ Module	Planned Coverage	Count	Priority
1.	Login and Authentication	Login/logout, invalid credentials, required-field checks, and session behavior	15	Critical
2.	Dashboard and Navigation	Dashboard widgets, menus, quick-launch panels, and navigation links	15	High
3.	PIM Module	Employee creation, editing, deletion, and employee search	15	Critical
4.	Leave Module	Leave request, leave approval/rejection, leave balance checks	15	High

5.	Time Module	Attendance tracking, timesheets, punch-in/punch-out validation	15	High
6.	Recruitment Module	Vacancies, candidate management, interview scheduling	15	Medium
7.	Admin Module	User management, role assignment, permission handling	15	Critical
8.	My Info and Support	Personal information updates, password changes, support/help pages links	15	Medium

Detailed Module Coverage

Login and Authentication

- Check mandatory-field warnings when login inputs are left blank.
- Verify login using valid admin credentials.
- Confirm invalid credentials show proper error messages.
- Validate logout functionality and session cleanup.
- Verify unauthorized users cannot access protected pages directly.

Dashboard and Navigation

- Verify dashboard widgets load properly after login.
- Confirm quick-launch panels redirect correctly.
- Validate sidebar and top-menu navigation.
- Check responsiveness of navigation links.
- Verify breadcrumb navigation consistency.

PIM Module

- Verify employee records can be added successfully.
- Confirm mandatory validation while creating employees.
- Validate employee search functionality using ID and name.
- Check edit and delete employee operations.
- Verify employee details persist after updates.

Leave Module

- Verify leave requests can be submitted successfully.
- Confirm leave balances update correctly.
- Validate approval and rejection workflows.
- Check date validation for leave applications.
- Verify leave status updates properly.

Time Module

- Verify attendance punch-in and punch-out functionality.
- Confirm timesheet entries can be created and edited.
- Validate total work-hour calculations.
- Check attendance records displayed correctly.
- Verify time entries persist after saving.

Recruitment Module

- Verify job vacancies can be added successfully.
- Confirm candidate applications are recorded correctly.
- Validate interview scheduling functionality.
- Check candidate search and filtering.
- Verify recruitment status updates display correctly.

Admin Module

- Verify new system users can be created.
- Confirm user-role assignments work correctly.
- Validate search functionality for system users.
- Check account enable/disable functionality.
- Verify admin permission restrictions behave correctly.

My Info and Support

- Verify personal details can be updated successfully.
- Confirm password-change validation works correctly.
- Check profile page navigation consistency.
- Validate emergency-contact and dependent information updates.
- Verify support/help pages load correctly.

Types of testing planned

Testing Type	Purpose in the capstone
Validation	Check required fields, incorrect formats, and empty submissions.
Negative	Confirm that incorrect or incomplete actions are handled safely.
Functional	Confirm HR workflows behave as expected from the user perspective
Navigation	Check menus, links, page transitions, and account movement.
Regression	Re-run important flows after script or data changes.
UI assertion	Verify visible headings, labels, counts, statuses, and confirmation messages

Automation approach

The automation layer will be built with Playwright Test and kept modular so each service can run independently. The intent is to keep the suite readable, maintainable, and easy to expand later without reworking the whole project.

- Use Playwright Test as the runner and keep each module in a separate spec file.
- Prefer stable locators such as role, label, placeholder, and visible text.
- Keep reusable setup logic small and focused, especially for authentication and employee setup.
- Store dummy data separately from the test logic so employee names and credentials can be changed quickly.
- Use assertions for URLs, visible messages, employee records, leave statuses, and attendance details.

Test strategy

- All inputs will use dummy employee and HR-related data.
- Unique usernames or employee IDs should be generated where necessary.
- Test scenarios should focus on visible OrangeHRM modules and workflows.
- Any required leave, attendance, or recruitment data should be reusable and easy to refresh.
- If the site changes its visible data, assertions should be updated without changing the overall testing intent..

Reporting and evidence

- The Playwright HTML report will act as the primary execution record.
- Screenshots should be saved for important failures or proof of execution.
- Trace files can help explain unexpected behavior and make debugging easier.
- The final summary should show planned tests, automated tests, pass count, fail count, and pending items.

Planned capstone flow

1. Finalize the website choice and testing scope.
2. Prepare 120 test cases across eight services.
3. Review and refine cases based on feedback.
4. Set up the Playwright project structure.
5. Automate the service-wise flows.
6. Run the suite, inspect failures, and fix issues.
7. Package the report, evidence, and submission folder.

Final Deliverables

Deliverables	Description
Capstone testing plan	This revised document covers scope, services, approach, and expected output.
Test case document	A service-wise set of 120 test cases with steps, expected results, priority, and status.
Playwright automation project	JavaScript test scripts organized by module
Test data files	Dummy employee, login, leave, recruitment, and attendance data
Execution report	The HTML report produced after the test run.
Evidence folder	Screenshots, traces, or notes for important passed/failed scenarios.
Final summary	A concise wrap-up with coverage, results, known issues, and conclusion.

Conclusion

This OrangeHRM functional testing plan provides a clear and practical approach to validating core HR management functionalities such as authentication, employee management, leave handling, recruitment, attendance tracking, and administrative workflows. With structured test coverage and Playwright automation, the project demonstrates effective QA practices, automation design, and a strong understanding of enterprise-level testing workflows.